

# SACM/CT Study of the dissociation/recombination dynamics of hydrogen peroxide on an *ab initio* potential energy surface

## Part II. Specific rate constants $k(E, J)$ , thermal rate constants $k_{\infty}(T)$ , and lifetime distributions†

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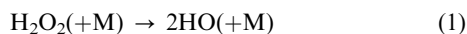
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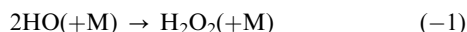
Statistical adiabatic channel model/classical trajectory (SACM/CT) calculations of the dissociation/recombination dynamics of hydrogen peroxide,  $\text{H}_2\text{O}_2 \rightleftharpoons 2\text{HO}$ , have been performed on an *ab initio* potential energy surface by Kuhn, Rizzo, Luckhaus, Quack, and Suhm (*J. Chem. Phys.* 1999, **111**, 2565). Specific rate constants  $k(E, J)$ , thermal rate constants  $k_{\infty}(T)$ , and lifetime distributions are determined. After averaging over  $J$ , the derived  $k(E, J)$  are in quantitative agreement with non-exponential time-profiles of HO formation recorded after overtone excitation of  $\text{H}_2\text{O}_2$  near the dissociation threshold by Scherer and Zewail (*J. Chem. Phys.* 1987, **87**, 97). The thermal high pressure rate constants for HO recombination agree with experimental data as well and can be represented by  $k_{\text{rec}, \infty}/10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \approx [0.376 (298 \text{ K}/T)^{0.47} + 0.013 (T/298 \text{ K})^{0.74}]$  over the range 60–1500 K. Non-statistical lifetime distributions are suggested not to have been of major relevance for the available experiment.

### 1. Introduction

The dissociation of  $\text{H}_2\text{O}_2$



and the reverse recombination



are bond fission/bond formation processes which have been studied frequently in theoretical modelling and detailed experimental work. On the one hand, the rates of the thermal dissociation<sup>1–14</sup> and thermal recombination<sup>15–18</sup> have been measured over wide temperature and pressure ranges in a variety of bath gases M. In addition, high overtone excitation experiments provided information of state-resolved dissociation rates and product energy distributions under collision-free conditions; at the same time, the vibrational spectroscopy of  $\text{H}_2\text{O}_2$  has been elaborated up to high excitation energies.<sup>19–34</sup>

On the other hand, the opportunities for theoretical modelling appear particularly challenging. The potential is complicated with a minimum energy path showing an intermediate “reef”, *i.e.* a maximum below the energy of separated HO radicals, separating valence-bound  $\text{H}_2\text{O}_2$  from hydrogen-bonded HOHO which finally dissociates into two dipolar HO.<sup>35–46</sup> It appears of interest to investigate whether the special properties of the potential are kinetically relevant. One also may ask about the role of excited electronic states which interact with the electronic ground state at large

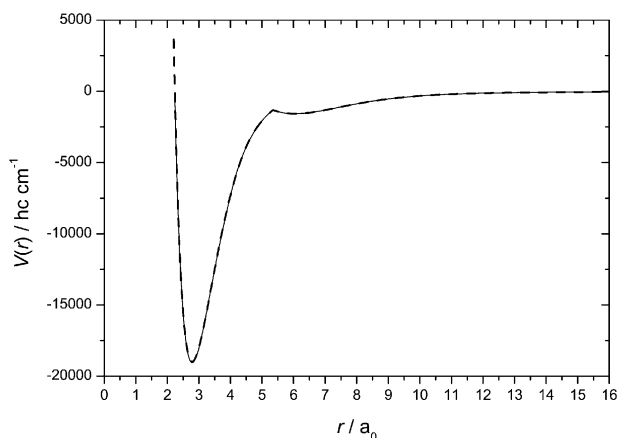
HO–HO distances.<sup>41,42</sup> The dynamics of the bond fission/bond formation process can be considered on different levels. At energies very close to the dissociation threshold, which are relevant for the recombination at very low temperatures, the statistical adiabatic channel model (SACM) has been applied on a very detailed level,<sup>47</sup> fully accounting for the subtle coupling of rotational, orbital, electronic orbital, and spin angular momenta. Classical trajectory calculations become appropriate for larger energies; they neglect such quantum effects but allow for a detailed classical characterization of intramolecular mode couplings.<sup>38–40,48–54</sup> Simplified SACM calculations<sup>55</sup> provided a semiquantitative rationalization of experimental specific rate constants  $k(E, J)$ . An important element of these calculations are adequate anharmonic densities of states such as calculated, at least for small energies, on *ab initio* potentials in ref. 43, 56–58. Thermal unimolecular rate theory led to simplified representations of falloff curves for the dissociation and recombination reactions.<sup>14,16–18</sup>

The opportunities for a rigorous comparison of theoretical modelling of the dissociation/recombination dynamics on the *ab initio* potential and the experimental results have not been exploited so far. Since we have recently extended to higher pressures earlier experimental studies of the low pressure range of the dissociation<sup>14</sup> (part I of this series), complementing earlier studies of the falloff curves of the recombination,<sup>17,18</sup> we became motivated to return to the theoretical modelling. This is the subject of the present work. We analyze the influence of the complicated properties of the potential on the various quantities relevant for the kinetics, such as centrifugal barriers, rovibrational numbers of open channels, and rovibrational anharmonic densities of states. In addition, as the dissociation experiments seemed to provide evidence for an influence of nonexponential lifetime distributions on the shape

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**Fig. 1** Minimum energy path (MEP) potential  $V(r)$  for the reaction  $\text{H}_2\text{O}_2 \rightleftharpoons 2\text{HO}$  ( $r$  = center-of-mass distance between the HO,  $a_0 = 0.529 \text{ \AA}$ , full line = *ab initio* results from ref. 45, dashed line = analytical fit by eqns (2)–(4) from this work).

of falloff curves, as discussed in ref. 59, our present work also devotes some attention to this aspect of the dynamics.

## 2. Potential energy surface

*Ab initio* calculations of the potential at large HO–HO distances, such as most relevant for the dissociation/recombination kinetics, have been pioneered in ref. 41 and 42. As we need analytical expressions for the global potential, we used the more detailed *ab initio* results from ref. 45 and their analytical representation. First, from this potential we determined the minimum energy path (MEP). The results could well be fitted in the form

$$V(r)/\text{hc cm}^{-1} = -192400 F(r)(r/a_0)^{-3} \quad (2)$$

with

$$F(r) = 1 - [-560.9 + 345.2 (r/a_0) - 75.22 (r/a_0)^2 + 7.183 (r/a_0)^3 - 0.3123 (r/a_0)^4 + 0.00515 (r/a_0)^5] \exp(-0.4435 r/a_0) \quad (3)$$

for  $r > r_c$ , and

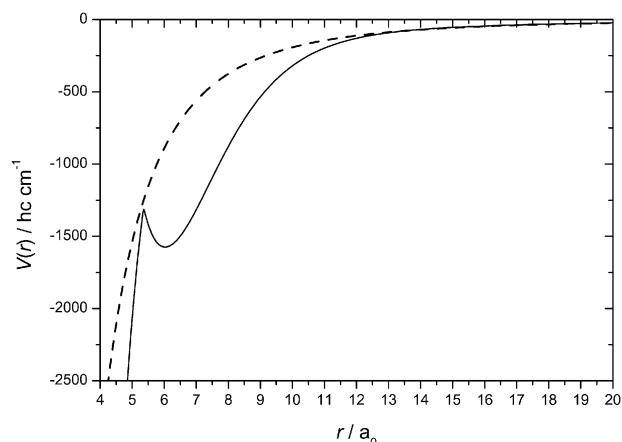
$$F(r) = -93.06 + 646.1 (r/5a_0) - 1968 (r/5a_0)^2 + 3433 (r/5a_0)^3 - 3512 (r/5a_0)^4 + 1938 (r/5a_0)^5 - 442 (r/5a_0)^6 \quad (4)$$

for  $r < r_c$ , where  $a_0 = 0.5292 \text{ \AA}$ ,  $r_c = 5.339 a_0$ , and  $r$  = center-of-mass distance between the two HO radicals;  $r_c$  denotes the  $r$ -value of the intermediate potential barrier (*i.e.*, of the reef). With eqn (2),  $V(r = r_c)/\text{hc} = -1312 \text{ cm}^{-1}$  corresponds to the energy of the reef of the potential. The limiting dipole–dipole potential from eqns (2) and (3) follows as

$$V_{\text{dd}}(r)/\text{hc cm}^{-1} = -192400(r/a_0)^{-3}. \quad (5)$$

Fig. 1 demonstrates that the present analytical MEP representation coincides very well with the detailed results from ref. 45; Fig. 2 illustrates the MEP of the dipole–dipole potential in comparison to that given in Fig. 1. One should note that the O–H distances are kept constant at  $r_{\text{O-H}} = 1.82 a_0$ .

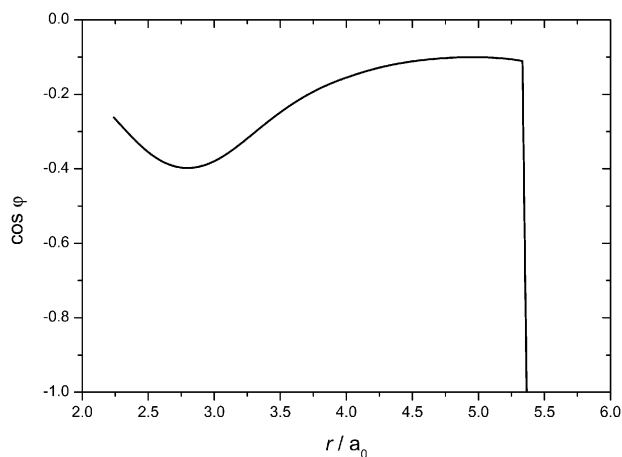
The angular dependence of the potential, like in our SACM/CT (SACM/classical trajectory) treatment of capture



**Fig. 2** As Fig. 1 (dashed line = dipole–dipole potential of eqn (5)).

dynamics between two linear rotors,<sup>60</sup> was represented as a function of the angles  $\theta_1$ ,  $\theta_2$ , and  $\varphi$ , where  $\theta_1$  and  $\theta_2$  denote the angles between the line connecting the two O atoms and the two HO axes while  $\varphi$  is the torsional angle. For details of the analytical representation of the angular dependence, the reader is referred to ref. 45. At this place, in Fig. 3 we only illustrate the geometry of the MEP by its respective  $r$ -dependence of  $\cos \varphi$ . The marked change of the character of the potential in the vicinity of the potential reef at  $r = r_c$  is illustrated in all figures. One should note that the potential becomes even more complicated when the interaction with excited electronic states<sup>41,42</sup> is taken into account, such as done in the SACM treatment for  $T \rightarrow 0 \text{ K}$ .<sup>27</sup> It is neglected in the present work, which concerns higher temperatures and where only reaction on the lowest potential needs to be considered, see below.

We explicitly specify the energy of the MEP in eqns (2)–(4), because our kinetics calculations are made on three levels: (i) by phase space theory (PST) neglecting centrifugal barriers and the anisotropy of the potential; (ii) by PST accounting for the centrifugal barriers  $E_0(J)$  such as calculated with  $V(r)$  of the MEP from eqns (2) and (3) but neglecting the anisotropy; and (iii) by introducing “rigidity”, *i.e.* accounting for the anisotropy of the potential. We have shown before that this



**Fig. 3** Angular dependence of the MEP potential, illustrated by the torsional angle  $\varphi$  (*ab initio* results from ref. 45).

strategy provides a particularly rational and transparent approach to the rate constants.

### 3. Centrifugal barriers

The calculation of centrifugal barriers  $E_0(J)$  forms an essential element of our analysis. These barriers are derived with the MEP potential  $V(r)$  as a function of the c.o.m.-distance  $r$ , such as represented by eqn (2), and the centrifugal energy  $V_{\text{cent}}(r) = \hbar^2 l(l+1)/2\mu r^2$  ( $l$  is the quantum number of orbital angular momentum). At this stage, we identify  $J$  (the quantum number of total angular momentum) with  $l$  which is appropriate for large values of the two quantum numbers. We first consider  $E_0(J)$  for the dipole-dipole potential of eqn (5). Here one obtains

$$E_{0,\text{dip-dip}}(J)/\text{hc} = 4[BJ(J+1)]^3/[(2M)^3 27C^2] \quad (6)$$

with  $C = 192\,400\text{ cm}^{-1}$ ,  $B$  = rotational constant of HO ( $18.87\text{ cm}^{-1}$ ), and  $M$  = effective mass of the system  $= \mu B a_0^2/\hbar^2 = 1.333$ . If centrifugal barriers were located at the position of the potential reef at  $r = r_c$ , one would have approximately

$$E_0(J) \approx V(r = r_c) + B\text{hc}J(J+1)/2[1 + M(r_c/a_0)^2]. \quad (7)$$

The true  $E_0(J)$  at small  $J$  is approximated by eqn (6), then switches to eqn (7), and finally deviates from eqn (7). We represent the complete  $E_0(J)$  by the maximum value  $J_{\text{max}}$  of  $J$  which, at a given energy  $E = E_0(J)$ , still permits to overcome the centrifugal barriers. Fig. 4 shows that the centrifugal barriers are always smaller than for the dipole-dipole potential which is in accord with the properties of Fig. 1. An analytical general representation of  $E_0(J)$  is given by

$$E_0(J) = E_{0,\text{dip-dip}}(J)\exp(y) \quad (8)$$

with

$$y = 0.2102[1 + \tanh(4z - 15.26)] - [1.442 - 0.2961z/4 - 0.8405(z/4)^2]\exp[6(z - 4)] \quad (9)$$

where  $z = \ln J$  and  $\ln J < 4.36$  (i.e.  $J < 78$ ), or

$$y = -1.31 - 3.193z - 1.13z^2 + 1.054z^3 + 8.663z^4 + 12.92z^5 - 9.74z^6 \quad (10)$$

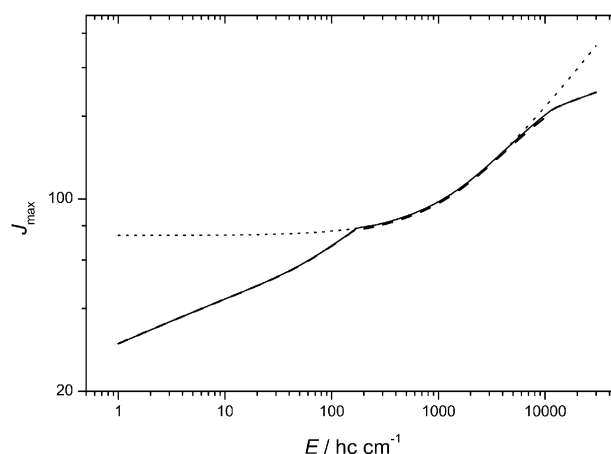
where  $z = \ln J - 5$  and  $\ln J > 4.36$  (i.e.  $J > 78$ ).

The ranges governed by the dipole-dipole potential and by the potential reef are well illustrated in Fig. 4. It should be emphasized that  $E_0(l)$  and  $E_0(J)$  differ at large energies when  $J$  is not much larger than the sum of rotational quantum numbers  $j_1$  and  $j_2$  of the two HO radicals. In this case, only  $E_0(l)$  or  $E_0(J, j_1, j_2)$  is defined. We will not consider further this range here.

### 4. Phase space theory for numbers of open channels $W^{\text{PST}}(E, J)$

Neglecting the centrifugal barriers  $E_0(J)$  as well as the anisotropy of the potential, the numbers of open channels  $W_0(E, J)$  of the system are obtained from

$$W_0(E, J) = \sum_{j_1} \sum_{j_2} \sum_j \sum_l 1 \quad (11)$$



**Fig. 4** Maximum  $J$ -values  $J_{\text{max}}$  allowing for overcoming the centrifugal barriers at a total energy of  $E$  ( $E(J_{\text{max}})$  defines the centrifugal barriers  $E_0(J)$ , full line = results for *ab initio* potential, dotted line = results for dipole-dipole potential, dashed line = results for centrifugal maxima above the potential reef at  $r = r_c$ ).

with the constraints  $0 \leq B\text{hc}j_1(j_1+1) \leq E$ ,  $0 \leq B\text{hc}j_2(j_2+1) \leq E - B\text{hc}j_1(j_1+1)$ ,  $|j_1 - j_2| \leq j \leq j_1 + j_2$ , and  $|J - j| \leq l \leq J + j$  where  $j$  corresponds to the combined rotational angular momenta of the two HO radicals. The limiting expressions of  $W_0(E, J)$  for small  $J$  and large  $J$  are well known<sup>61–63</sup> and interpolations between the two limiting expressions have been described in ref. 62–65. Here, we have gone beyond the interpolations by accurately calculating  $W_0(E, J)$  from eqn (11) which, in the classical limit, gives

$$W_0(E, J) = J^4/6 + 8J_0^3J/3 - (J^3 + 13J_0^2J)(J_1^2 - J^2)^{1/2}/12 - (J^2J_0^2 + J_0^4/2)\arctan[J/(J_1^2 - J^2)^{1/2}] \quad (12)$$

for  $J < J_0$ ,

$$W_0(E, J) = J_0^4/2 + (J^3 + 13J_0^2J)(J_1^2 - J^2)^{1/2}/12 - (J^2J_0^2 + J_0^4/2)\arctan[J/(J_1^2 - J^2)^{1/2}/J] \quad (13)$$

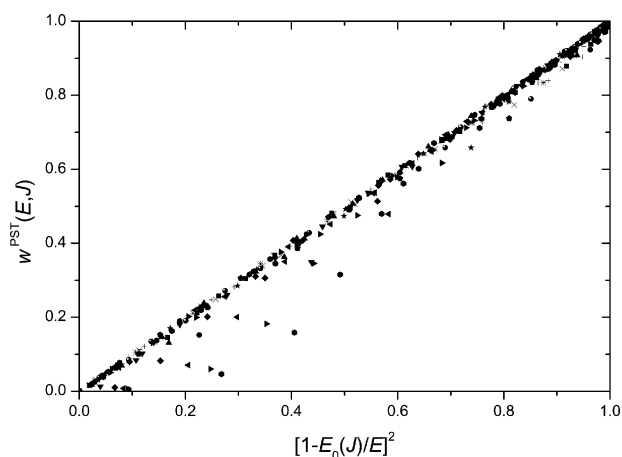
for  $J_0 < J < J_1$ , and

$$W_0(E, J) = J_0^4/2 \quad (14)$$

for  $J > J_1$ , with  $J_0 = (E/B\text{hc})^{1/2}$ , and  $J_1 = (2E/B\text{hc})^{1/2}$ . One should note that eqns (12)–(14) correspond to two non-identical rotors (having, however, identical rotational constants). In the present case of two identical rotors,  $W_0(E, J)$  is divided by 2 in its following use.

Eqns (12)–(14) represent our first approximation to the true  $W(E, J)$ . In the second approximation, we account for the centrifugal barriers, but still neglect the anisotropy of the potential. This approximation corresponds to phase space theory (PST) on the true MEP potential. The ratio  $W^{\text{PST}}(E, J)/W_0(E, J)$  represents the capture probability in PST,  $w^{\text{PST}}(E, J)$ , such as discussed in ref. 66 for dipole-dipole capture. Similar to the results from ref. 66,  $w^{\text{PST}}(E, J)$  can well be approximated by

$$w^{\text{PST}}(E, J) \approx [1 - E_0(J)/E]^2, \quad (15)$$



**Fig. 5** Capture probabilities  $w^{\text{PST}}(E, J)$  for isotropic potential, *i.e.* in phase space theory (PST) ( $E_0(J)$  from Fig. 4, CT calculations for  $E/hc \text{ cm}^{-1} = 1$  ( $\square$ ), 2.9 ( $\circ$ ), 8.3 ( $\triangle$ ), 24 ( $\nabla$ ), 70 ( $\diamond$ ), 200 ( $\triangleleft$ ), 335 ( $\blacksquare$ ), 570 ( $\bullet$ ), 1650 ( $\blacktriangle$ ), 4700 ( $\blacktriangledown$ ), 13600 ( $\blacklozenge$ ), 30000 ( $\times$ )).

which is illustrated in Fig. 5. The points in Fig. 5 were obtained by numerically evaluating eqn (11) with and without centrifugal barrier constraints. At the same time capture probabilities  $w^{\text{PST}}(E, J)$  were determined by classical trajectory calculations (for details see ref. 66), leading to identical results. Only at very high energies and when  $J \approx l$  is not fulfilled, some deviations and scatter are observed, as seen in Fig. 5.

### 5. Specific rigidity factors $f_{\text{rigid}}(E, J)$

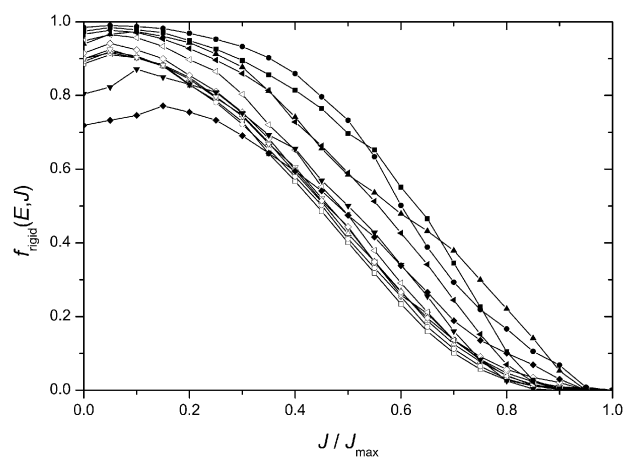
The anisotropy of the potential reduces the numbers of open channels and the corresponding capture probabilities  $w(E, J)$  to values below the PST results. The reduction factor  $W(E, J)/W^{\text{PST}}(E, J) = w(E, J)/w^{\text{PST}}(E, J)$  is termed the specific rigidity factor  $f_{\text{rigid}}(E, J)$ .<sup>66</sup> We have determined  $w(E, J)$  and  $f_{\text{rigid}}(E, J)$  by classical trajectory calculations for the capture of two HO radicals forming  $\text{H}_2\text{O}_2$  on the *ab initio* potential, including and excluding the anisotropy of the potential. For details of our trajectory calculations and the criteria for capture, see ref. 66. It is convenient to represent  $f_{\text{rigid}}(E, J)$  as a function of  $J/J_{\text{max}}(E)$  where  $J_{\text{max}}(E)$  is shown in Fig. 4. Fig. 6 gives the results. The spread of the curves with varying energies is not very pronounced. Only at very high energies and small  $J$ , major deviations from the general trends occur. The relation

$$f_{\text{rigid}}(E, J) \approx \{1 - [J/J_{\text{max}}(E)]^2\}^3 \quad (16)$$

fairly well represents all data for  $E/hc \leq 350 \text{ cm}^{-1}$ . This is the result also obtained in ref. 66 for dipole-dipole capture. At higher energies, there is an intermediate range, where

$$f_{\text{rigid}}(E, J) \approx 1 - [J/J_{\text{max}}(E)]^2 \quad (17)$$

provides a better approximation, before  $f_{\text{rigid}}(E, J = 0)$  with increasing energy decreases below unity. The switching between the ranges is a consequence of the complicated anisotropy of the potential. In our work the true  $f_{\text{rigid}}(E, J)$ , as represented in Fig. 6, without approximations is used for the subsequent calculations of dynamic quantities.



**Fig. 6** Specific rigidity factors  $f_{\text{rigid}}(E, J) = w(E, J)/w^{\text{PST}}(E, J)$  (CT calculations as in Fig. 5).

### 6. Thermal capture rate constants

The thermal capture rate constants  $k_{\text{cap}}(T)$ , which lead to high pressure recombination and the reverse dissociation rate constants such as elaborated below, follow directly from the numbers of open channels  $W(E, J) = W^{\text{PST}}(E, J) f_{\text{rigid}}(E, J)$  of sections 4 and 5 through

$$k_{\text{cap}}(T) = \frac{kT}{h} \left( \frac{h^2}{2\pi\mu kT} \right)^{3/2} \sum_{J=0}^{\infty} (2J+1) \int_0^{\infty} W(E, J) \times \exp\left(-\frac{E}{kT}\right) \frac{dE}{kT} \quad (18)$$

$k_{\text{cap}}(T)$  can either be obtained by thermal averaging of  $W(E, J)$  from sections 4 and 5 or by direct trajectory calculations with thermal sampling of initial conditions (for details see ref. 60 and 66). We again applied the two methods and obtained identical results, confirming the internal consistency of our treatment. Fig. 7 shows the results. At  $T \leq 15 \text{ K}$ , the pure dipole-dipole capture rate constants are obtained, as given by ref. 60

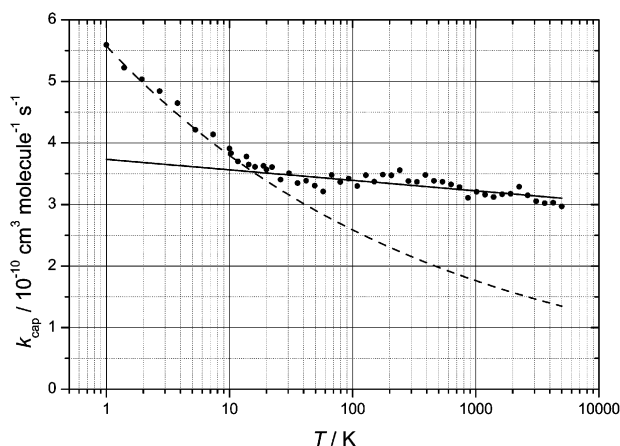
$$\begin{aligned} k_{\text{cap, dip-dip}} &= (1/2)0.354\Gamma(1/3)\mu_D^{4/3}(kT)^{-1/6}\sqrt{8\pi/\mu} \\ &= 2.16(298 \text{ K}/T)^{1/6} \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \end{aligned} \quad (19)$$

The stoichiometric factor 1/2 in eqn (19) corresponds to the capture of two identical dipolar rotors;  $\mu_D$  denotes the dipole moment of HO radicals, see Appendix. At 15 K, there is an abrupt switch to a range where the potential reef at  $r = r_c$  governs the capture rate. Above 15 K,  $k_{\text{cap}}$  can then well be represented by

$$k_{\text{cap}} \approx [3.31 - 0.074 \ln(T/298 \text{ K})] \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}. \quad (20)$$

Fig. 7 compares the trajectory points with eqns (19) and (20).

Comparing Fig. 7 with the corresponding PST results leads to the thermal rigidity factors.<sup>60</sup> Thermal averaging of specific rigidity factors and direct trajectory results from thermally sampled initial conditions agree very well. At low



**Fig. 7** Thermal capture rate constants  $k_{\text{cap}}(T)$  for  $2\text{HO} \rightarrow \text{H}_2\text{O}_2$  (points = CT calculations on *ab initio* potential, dashed line = CT calculations on dipole–dipole potential, full line = representation by eqn (20)).

temperatures, the thermal rigidity factor of pure dipole–dipole capture, *i.e.*  $f_{\text{rigid}} = 0.354$ ,<sup>60</sup> is correctly recovered. In spite of the marked differences between  $k_{\text{cap}}(T)$  and  $k_{\text{cap,dip-dip}}(T)$  illustrated in Fig. 7, the thermal rigidity factors  $f_{\text{rigid}}(T)$  do not differ too much from the value for dipole–dipole capture. This may appear surprising but it signals that the anisotropy character of the potential does not change too much in the range of  $r$  near  $r_c$ .

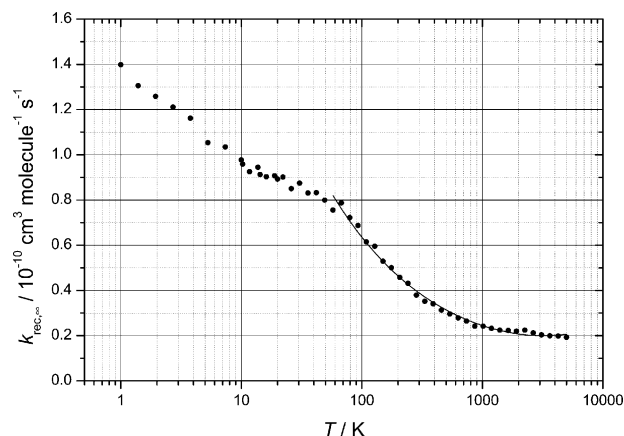
In order to derive thermal high pressure recombination rate constants  $k_{\text{rec},\infty}(T)$  from  $k_{\text{cap}}(T)$ , one must have information on the dynamic interactions between the electronic states of  $\text{H}_2\text{O}_2$  correlating with the fine structure components  $\text{HO}(\text{}^2\Pi_{3/2})$  and  $\text{HO}(\text{}^2\Pi_{1/2})$ , see ref. 42. This is not easily obtained. If, for simplicity, we assume that only the lowest potential arising from the electronic fine structure component  $\text{HO}(\text{}^2\Pi_{3/2})$  leads to recombination, the high pressure recombination rate constant  $k_{\text{rec},\infty}$  of reaction (–1) follows from  $k_{\text{cap}}$  by division through the square of the electronic partition function of HO, *i.e.*

$$k_{\text{rec},\infty} = k_{\text{cap}}[2 + 2 \exp(-201 \text{ K}/T)]^{-2}. \quad (21)$$

Fig. 8 shows the results which, over the range 60–5000 K, can be approximated by

$$k_{\text{rec},\infty} \approx [0.376(298 \text{ K}/T)^{0.47} + 0.013(T/298 \text{ K})]^{0.74} \times 10^{-10} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}. \quad (22)$$

The calculated value of  $k_{\text{rec},\infty}(298 \text{ K}) \approx 3.89 \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$  is in fair agreement with the experimental value of  $(2.6 \pm 0.8) \times 10^{-11} \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1}$  from ref. 17, which is obtained by a long and somewhat problematic (see below) falloff extrapolation and which is also influenced by the problems of separating reaction (–1) from the competing process  $2\text{HO} \rightarrow \text{H}_2\text{O} + \text{O}$ .<sup>15–18</sup> In any case, the calculated value is not below the experimental value. The opposite behaviour would have indicated contributions from higher electronic states. The agreement between experiment and theory then confirms our assumption that only the lowest of the potential energy surfaces of  $\text{H}_2\text{O}_2$ , arising from the lowest



**Fig. 8** Thermal high pressure rate constants  $k_{\text{rec},\infty}(T)$  for  $2\text{HO} \rightarrow \text{H}_2\text{O}_2$  (points = CT calculations on the *ab initio* potential, assuming reaction only on the potential arising from the lowest fine structure components of HO ( $\text{}^2\Pi_{3/2}$ ), line = representation by eqn (22)).

fine structure component of electronic ground state HO radicals, contributes to the recombination reaction.

While the agreement between recombination experiments and theory appears quite satisfactory near room temperature, simple falloff extrapolations of high temperature dissociation experiments near 1000 K from ref. 14 led to values of  $k_{\text{rec},\infty}$  which are about one order of magnitude lower than those calculated here. The present calculations of  $k_{\text{rec},\infty}$ , therefore, may be used to construct more correct falloff curves of the system at higher temperatures. This will be the aim of a future publication, see below. Rate constants for temperatures below 50 K, for which quantum effects of various types and coupling between several electronic states need to be considered,<sup>67–71</sup> require more detailed quantum-chemical investigations which go beyond the scope of the present article.

## 7. Specific rate constants $k(E,J)$

In order to compare our calculations with experimental data for specific rate constants of  $\text{H}_2\text{O}_2$  dissociation<sup>25,26,28,34</sup> and with experimental falloff curves for dissociation and recombination, beyond  $W(E,J)$  also rovibrational densities of states  $\rho(E,J)$  of  $\text{H}_2\text{O}_2$  have to be constructed. In statistical unimolecular rate theory, one then has

$$k(E,J) = W(E,J)/\text{hp}(E,J) \quad (23)$$

Because of the complicated potential of  $\text{H}_2\text{O}_2$ , with a hindered internal rotor/torsional mode near the equilibrium configuration and the special minimum energy path potential of Fig. 1 at higher energies, the adequate calculation of  $\rho(E,J)$  is not simple. On the other hand, accurate calculations of level positions and hence  $\rho(E,J = 0)$  have been made in ref. 43, 56–58 up to relatively high energies. These were, however, still far below the energies of dissociating  $\text{H}_2\text{O}_2$ . We have treated the problem of calculating the highly anharmonic  $\rho(E,J)$  in various ways: (i) we have expressed a quasi-harmonic density of states  $\rho(E,J = 0)$  by the Whitten-Rabinovitch equation<sup>72</sup> with the harmonic frequencies given in the Appendix. One frequency, corresponding to the torsion, was arbitrarily

chosen in such a way that the Whitten-Rabinovitch expression

$$\rho(E, J=0) = (E + a(E)E_z)^5 / 5! \prod_{i=1}^6 (h\nu_i) \quad (24)$$

at low energies well approximated the results from the accurate state calculations of ref. 43, 56–58 (one should note that, unlike the rest of the article,  $E$  in  $\rho(E, J)$  and  $k(E, J)$  is counted from the rovibrational ground state of  $\text{H}_2\text{O}_2$ ). (ii) We have used the accurate state calculations from ref. 43, 56–58, which are available up to energies of about  $8000 \text{ cm}^{-1}$ , and extrapolated them by the Whitten-Rabinovitch energy dependence, i.e., by

$$\rho(E, J=0)/\rho(8000 \text{ cm}^{-1}, J=0) \approx [(E + a(E)E_z)/(8000 \text{ hc cm}^{-1} + a(8000 \text{ cm}^{-1})E_z)]^5 \quad (25)$$

(iii) We have calculated  $\rho(E, J=0)$  by Monte Carlo integration of the phase space with the *ab initio* potential energy surface from ref. 45. At energies in the unstable range at  $E \geq E_0$ , the phase space of  $\text{H}_2\text{O}_2$  was limited by the positions of the centrifugal barriers of the orbital motion. As the phase space integration gives classical numbers or densities of states, the results had to be corrected for quantum effects. This was done, as described in ref. 73, by multiplying the classical  $\rho(E, J=0)$  by the factor  $[(E + a(E)E_z)/(E + E_z)]^5$ . (iv) We have also applied the anharmonicity model of ref. 74 with Morse anharmonicities in the stretching modes, harmonic bends with the empirical stretch-bend couplings from ref. 74, and a harmonic torsion. Fig. 9 compares the results from the various methods for calculating  $\rho(E, J=0)$ . In comparison to the harmonic density of states, as defined by method (iii), there is an anharmonicity factor

$$F_{\text{anh}}(E, J=0) \approx [E + a(E)E_z]/\text{hc } 17050 \text{ cm}^{-1} \quad (26)$$

for  $E/\text{hc} > 11400 \text{ cm}^{-1}$  and  $F_{\text{anh}}(E, J=0) \approx 1$  for smaller  $E$ -values. One should, however, note that a major part of the additional anharmonicity is minimized by the choice of an empirical torsion frequency, as given in the Appendix.

While methods (iii) and (iv) nearly agree in  $F_{\text{anh}}(E_0, J=0)$ , method (iv) at high energies slightly overestimates the anharmonicity contribution in comparison to the phase space integration. One might improve the result from method (iv) by the modifications tested in ref. 75. However, as we have the phase space integration, here we can directly use the results from method (iii). These can be expressed by eqns (24) and (26). ( $\rho(E, J=0)$  from eqn (24) for  $E \geq E_0$  can well be represented by  $\rho(E, J=0) = (5.97/\text{cm}^{-1})[(E + a(E_0)E_z)/(E_0 + a(E_0)E_z)]^5$  with  $a(E_0) \approx 0.95$ ,  $E_z/\text{hc} = 5690 \text{ cm}^{-1}$ , and  $E_0/\text{hc} = 17050 \text{ cm}^{-1}$ ). The accurate state calculations from method (ii) using the DVR method apparently cannot be extended far beyond  $8000 \text{ cm}^{-1}$  without a very large effort, such that the present approach gives more appropriate results for higher energies.

We have also explored the  $J$ -dependence of  $\rho(E, J)$  in two ways. (i) We have done phase space integrations for representative  $J$ -values. (ii) We have compared the results with the relationships for  $\rho(E, J)/\rho(E, J=0)$  elaborated in ref. 64 and 76 (for a prolate symmetric top molecule, given by eqns (2.8), (2.11)–(2.16) from ref. 64). Since the agreement between the

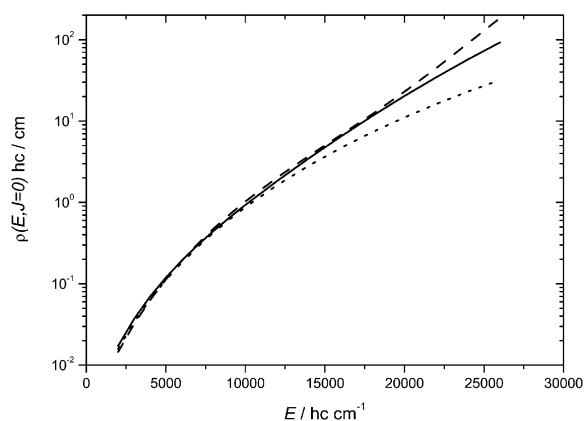


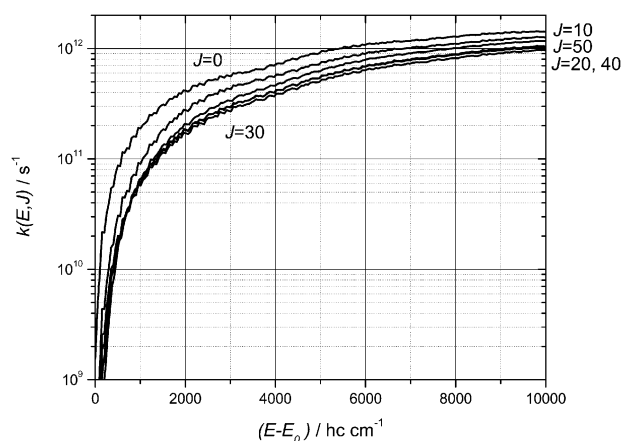
Fig. 9

**Fig. 9** Density of states  $\rho(E, J=0)$  for  $\text{H}_2\text{O}_2$  ( $E$  = energy above the vibrational ground state of  $\text{H}_2\text{O}_2$ ,  $E_z/\text{hc}$  being  $5690 \text{ cm}^{-1}$  above the potential minimum; dotted line = harmonic oscillator count with harmonic torsion frequency  $340 \text{ cm}^{-1}$ ; full line = Monte Carlo integration of phase space, quantized with eqn (24); dashed line = empirical anharmonicity model from ref. 74, see text).

two calculations in all cases was found to be excellent, we finally used eqns (24), (26) and  $\rho(E, J)/\rho(E, J=0)$  from ref. 64 for the representation of  $\rho(E, J)$ , as required in the calculation of the specific rate constants  $k(E, J)$ .

Combining  $W(E, J)$  from sections 3–5 with  $\rho(E, J)$  from the present section gives  $k(E, J)$ . Fig. 10 shows our results for  $J$  in the range 0 to 50 and for energies up to  $10000 \text{ cm}^{-1}$  above the dissociation energy. One recognizes a coarse structure due to the opening of channels with one or two quanta of HO stretching near  $3500$  and  $7000 \text{ cm}^{-1}$ . There is the additional fine structure due to the quantization of the transitional bending modes which we introduced by quantizing the rotational states of HO in the expressions for  $W_0(E, J)$ , see above.

The comparison of our calculated  $k(E, J)$  with experimentally determined specific rate constants is not as direct as one would wish. A first attempt to relate the overtone line width results from ref. 25 and the real-time measurements of HO production after overtone excitation from ref. 26 with

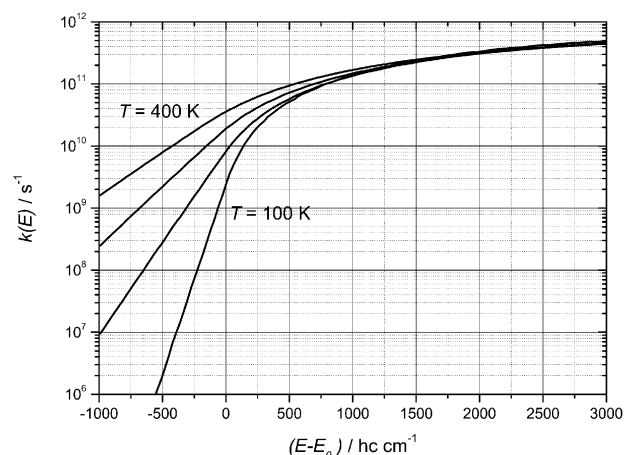


**Fig. 10** Specific rate constants  $k(E, J)$  for dissociation of  $\text{H}_2\text{O}_2$  (SACM/CT results on *ab initio* potential, curves from top to bottom for  $J = 0, 10, 50, 20$  and  $40$  (nearly coinciding), and  $30$ , see text;  $E$  = energy above the rovibrational ground state of  $\text{H}_2\text{O}_2$ ).

simplified SACM calculations of  $k(E, J)$ , on the basis of empirically calibrated potential parameters, was made in ref. 55. The consistent picture obtained appeared encouraging but should be considered with caution. It did not yet account for the details of the experiments and it used empirical fitted potential parameters in the SACM calculations. The results, which became available since the analysis from ref. 55, now allow for a more detailed comparison between experiments and theory. On the one hand, more detailed real-time measurements of HO production<sup>28</sup> and overtone line width analyses<sup>34</sup> have become available; on the other hand, the present SACM/CT calculations of  $k(E, J)$  on a parameter-free *ab initio* potential can be employed for the analysis. Nevertheless, there remain questions about the details of the excitation and probing conditions characterizing the experiments from ref. 28 and 34 which make the comparison with the calculations still somewhat tentative.

A detailed analysis of the various contributions to the structure and width of overtone bands and a comparison with SACM calculations of the simplified type described in ref. 55 was performed in ref. 4. After unravelling the various contributions to the line widths, it was concluded that  $k(E)$  should be a factor of 2.3 smaller at  $E/hc \approx 1900 \text{ cm}^{-1}$  than that proposed in ref. 55. The present parameter-free calculation of  $k(E, J)$  led to similar specific rate constants as in ref. 55. Thus, there remains a discrepancy by about a factor of 2 between the theoretical value and the value derived by modelling the experiments. One may attribute this discrepancy to uncertainties in the linewidth analysis, being determined by several superimposed contributions, or to non-statistical contributions to  $k(E, J)$  calculations.

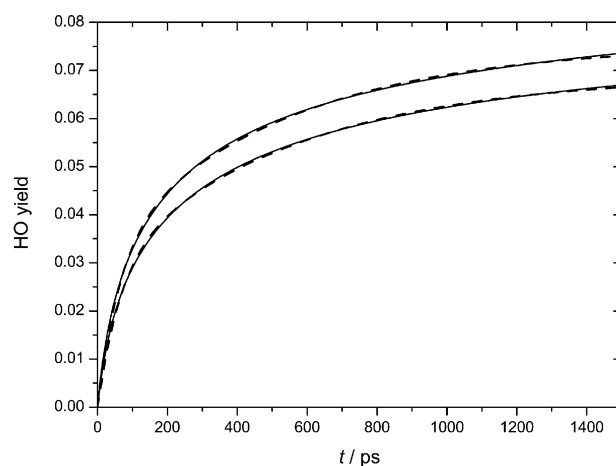
While the linewidth analysis of ref. 34 concerned excitation to energies about  $1900 \text{ cm}^{-1}$  above the dissociation threshold  $E_0$ , the real-time measurements from ref. 28 employed excitation energies below  $E_0$  such that only the high-energy tail of a thermal distribution reached into the dissociation range. As biexponential time-profiles of HO production were recorded, the question was discussed in ref. 28 whether this was the result of non-statistical dynamics or of a superposition of dissociations from different initial states. We have investigated this question with our calculated  $k(E, J)$ . We had to make some assumptions about the experimental excitation conditions, namely that neither the overtone excitation of the thermal  $\text{H}_2\text{O}_2$  ensemble nor the laser-induced fluorescence probing of the dissociation products HO lead to biased results with specific  $J$ -preferences. Based on these assumptions, Fig. 11 shows  $k(E)$  after averaging over thermal rovibrational distributions. At  $(E - E_0)/hc \approx 1900 \text{ cm}^{-1}$ , corresponding to the experiments from ref. 25 and 34,  $k(E) \approx 3.0 \times 10^{11} \text{ s}^{-1}$  is nearly independent of the temperature of the experiments but higher than the value from the analysis in ref. 34 which led to  $k(E) \approx 1.7 \times 10^{11} \text{ s}^{-1}$ . At  $(E - E_0)/hc \approx -765 \text{ cm}^{-1}$ , corresponding to most of the experiments from ref. 28,  $k(E) \approx 7 \times 10^8 \text{ s}^{-1}$  is calculated; for  $(E - E_0)/hc = -1072 \text{ cm}^{-1}$  which was also studied in ref. 28, we obtain  $1.6 \times 10^8 \text{ s}^{-1}$ . In both cases, ensembles of 300 K are considered. A detailed time-dependence of HO production, such as calculated with our  $k(E, J)$  for a thermal ensemble at 300 K, is shown in Fig. 12 for  $(E - E_0)/hc = -765 \text{ cm}^{-1}$ . One observes a bi-exponential



**Fig. 11** Average specific rate constants  $k(E)$  for canonical ensembles at temperature  $T$  ( $E$  = excitation energy excluding rovibrational energy, see text).

profile with the two time constants 63 and 540 ps. A PST calculation would have given quite similar results. One should also note that, for  $(E - E_0)/hc = -1072 \text{ cm}^{-1}$ , we obtained the quite similar time constants 65 and 600 ps. On the other hand, monoexponential time-profiles are obtained above the dissociation energy; e.g., for  $(E - E_0)/hc = +3000 \text{ cm}^{-1}$ , monoexponential HO formation with a time constants of 2.27 ps is obtained. The present modelling results, therefore, are in excellent agreement with the time-resolved experimental profiles from ref. 28. The bi-exponential profiles from ref. 28, as a consequence, cannot be used for conclusions on non-statistical dissociation dynamics. One should also note that the biexponential profiles modelled for  $E < E_0$  are characterized by more or less energy-independent time constants which, however, strongly depend on the temperature of the ensemble.

In summary, one can state that the calculated  $k(E, J)$ , within the ambiguities of characterizing the excitation conditions of



**Fig. 12** Modelled time dependence of HO formation in  $\text{H}_2\text{O}_2$  at 300 K and  $(E - E_0)/hc = -765 \text{ cm}^{-1}$  (lower curves: anisotropic potential, upper curves: PST; full lines: modelling results, dashed lines: bi-exponential approximation, lower curves:  $\text{HO yield}/0.069 = 0.41[1 - \exp(-t/\tau_1)] + 0.59[1 - \exp(-t/\tau_2)]$  with  $\tau_1 = 63 \text{ ps}$  and  $\tau_2 = 540 \text{ ps}$ ; upper curves:  $\text{HO yield}/0.075 = 0.43[1 - \exp(-t/\tau_1)] + 0.57[1 - \exp(-t/\tau_2)]$  with  $\tau_1 = 62 \text{ ps}$  and  $\tau_2 = 520 \text{ ps}$ ).

the experiments, are in good agreement with the measured quantities.

## 8. Thermal rate constants in the low pressure and the falloff region

The foregoing calculations of centrifugal barriers  $E_0(J)$  and of anharmonic rovibrational densities of states allow one to perform accurate calculations of the statistical factors in the limiting low pressure rate constants for recombination and dissociation,  $k_{\text{rec},0}$  and  $k_{\text{diss},0}$ , respectively. We use the formalism from ref. 77 with the contributing factors summarized in the Appendix. The calculation leads to strong collision rate constants for dissociation which can be represented by

$$k_{\text{diss},0}^{\text{SC}} = [\text{Ar}] 9.1 \times 10^{-7} \exp(-22\,740 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad (27)$$

over the range 800–1400 K. The comparison with the experimental values from ref. 14, 78 and 79, *i.e.*

$$k_{\text{diss},0}^{\text{exp}} = [\text{Ar}] 3.8 \times 10^{-8} \exp(-21\,960 \text{ K}/T) \text{ cm}^3 \text{ molecule}^{-1} \text{ s}^{-1} \quad (28)$$

obtained by some falloff extrapolation, gives fitted collision efficiencies  $\beta_c \approx 0.073$  (1200 K/ $T$ ). From that, average energies transferred per collision  $\langle \Delta E \rangle$  through

$$\beta_c / (1 - \beta_c^{1/2}) \approx -\langle \Delta E \rangle / E_{\text{F}} kT \quad (29)$$

of about  $\langle \Delta E \rangle / hc \approx -80 \text{ cm}^{-1}$  are derived. This value appears of reasonable magnitude if one compares it with values for the  $\text{HO}_2$  ( $-24 \text{ cm}^{-1}$ )<sup>80</sup> or  $\text{CH}_4$  ( $-50 \text{ cm}^{-1}$ )<sup>81</sup> systems, being evaluated in an analogous manner. However,  $\beta_c$  and  $\langle \Delta E \rangle$  are fit parameters only which may be somewhat modified by changes in the falloff extrapolations, see below, and need not to be identical for different reaction systems.

Our detailed foregoing calculations also allow us to determine more correct statistical strong collision falloff curves than used before, accounting for correct rotational effects and transitional mode properties, see ref. 82 and 83. Fig. 13 shows the results in the format of the broadening factors  $F^{\text{SC}}(k_0/k_\infty)$ . The nearly temperature independent broadening factors at the center  $F_c^{\text{SC}} \approx 0.5$ , for  $T = 100$ –300 K, correspond well to the rough estimate from ref. 84 (for the combination of two linear species  $F_c^{\text{SC}} \approx 0.5$  was recommended). Fig. 13 shows that symmetrical broadening factors, from the function

$$F^{\text{SC}}(x) \approx F_c^{\text{SC}} \{1 + [(\log x)/N]^2\}^{-1} \quad (30)$$

with  $x = k_0/k_\infty$  and  $N \approx 0.75$ – $1.27 \log F_c$ ,<sup>77</sup> differ from the present calculations to some extent. Some asymmetries are observed which have to be taken into account in comparisons with experiments. At this stage, it appears sufficient to represent the additional weak collision broadening effects by multiplying  $F_c^{\text{SC}}$  by the factor  $F_c^{\text{WC}} \approx \beta_c^{0.14}$  from ref. 77 which leads to the combined center broadening factors  $F_c$  (300 K)  $\approx 0.41$  and  $F_c$  (1200 K)  $\approx 0.44$  for the bath gas Ar.

Before deriving final experimental values of limiting low and high pressure rate constants employing the present detailed falloff curves, one should try to answer the question whether

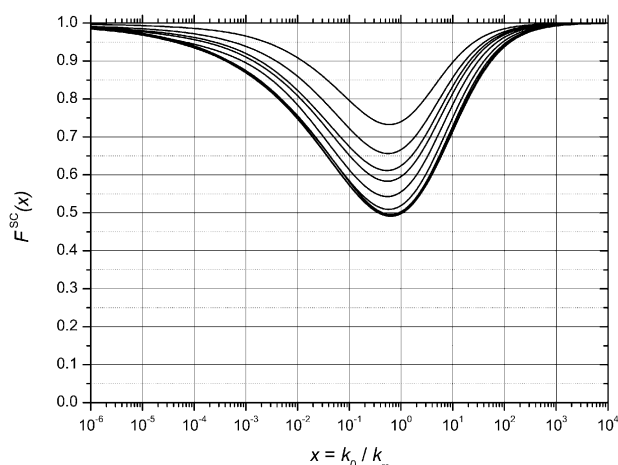


Fig. 13 Strong collision broadening factors  $F^{\text{SC}}(x)$  for  $2\text{HO} \rightleftharpoons \text{H}_2\text{O}_2$  (SACM/CT results for  $T/\text{K} = 100, 150, 200, 300, 500, 750, 1000, 1500, 2500$  from bottom to top, see text).

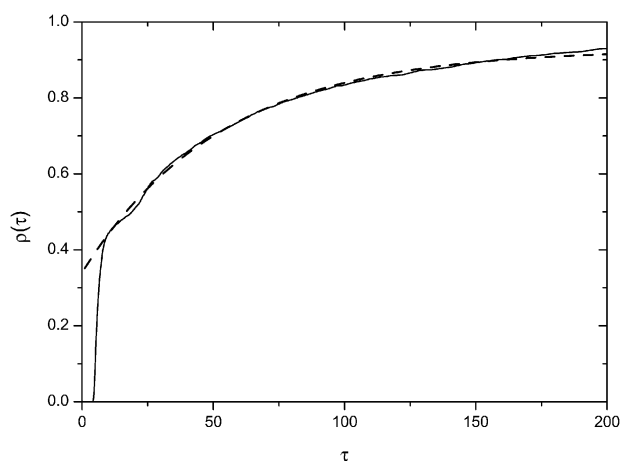
there may be additional falloff broadening effects from non-exponential, *i.e.* non-statistical, lifetime distributions of dissociating  $\text{H}_2\text{O}_2$ . This is done in the next section.

## 9. Lifetime distributions of dissociating $\text{H}_2\text{O}_2$

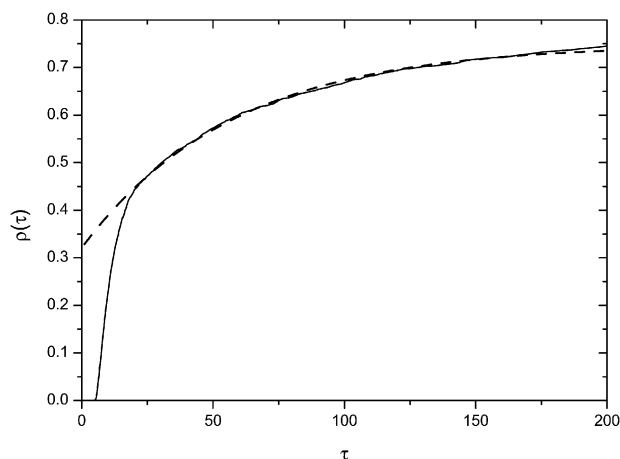
The question of non-statistical lifetime distributions of dissociating  $\text{H}_2\text{O}_2$  has been addressed in several previous studies, both from experimental viewpoint,<sup>14,28</sup> and from the position of trajectory modelling.<sup>38,39,50,51</sup> The previous CT studies employed artificial potentials such that their results do not appear to be of quantitative relevance although they produced  $k(E)$  of the correct magnitude.<sup>34,38,39</sup> In order to be more realistic, we have done CT lifetime calculations on the *ab initio* potential from ref. 45. We started our trajectories at c.o.m. distances of  $10a_0$  ( $a_0 = 0.529 \times 10^{-8} \text{ cm}$ ) between two approaching HO radicals and followed their trajectories until they separated again and reached  $10a_0$ . The starting ensembles were either chosen to be canonical at 300 K, or to be microcanonical with a given  $E$  and  $J$ . Fig. 14 and 15 show examples for cumulative distributions after microcanonical selection of initial conditions. There is a short time period of about 0.1 ps which corresponds to non-complex forming, direct, collisions. After that, the distribution is bi-exponential with two time constants being markedly different for large  $J$  (about a factor of 10) and much less different for  $J = 0$  (about a factor of 2). The cumulative distributions after canonical selection of initial conditions, on the other hand, are much closer to a multi-exponential distribution which can also be well approximated by a mono-exponential distribution. We, therefore, conclude that, although there may be some non-statistical lifetime distributions in this system, they will not have been detectable in the experiments of ref. 28. Instead, the apparent non-exponential profiles of HO-formation must almost exclusively have been caused by the rotational distribution in the observed ensembles.

## 10. Conclusions

The present SACM/CT treatment of the dissociation/recombination dynamics of  $\text{H}_2\text{O}_2$  on an *ab initio* potential energy



**Fig. 14** Cumulative lifetime distributions for  $\text{H}_2\text{O}_2$  dissociation (CT calculations on *ab initio* potential, see text;  $E/\text{hc} = 800 \text{ cm}^{-1}$ ,  $J = 0$ ,  $J_{\text{max}}(E) = 93$ ,  $\tau$  in units of  $4.32 \times 10^{-14} \text{ s}$ , full line = CT calculations, dashed line = exponential fit up to  $\tau = 200$  with time constant 2.28 ps, fit up to  $\tau = 2000$  with time constant 3.85 ps, see text).



**Fig. 15** As Fig. 14, but  $J = 56$ , (exponential fit up to  $\tau = 200$  with time constant 2.49 ps, fit up to  $\tau = 2000$  with time constant 22.2 ps, see text).

surface allowed us to calculate specific rate constants  $k(E, J)$  and high pressure rate constants  $k_{\infty}(T)$  for dissociation and recombination which are free from empirical fit parameters. Experimental time-profiles of HO production induced by overtone excitation from ref. 28 were quantitatively reproduced, indicating that the observed bi-exponential profiles can be completely accounted for by superposition of dissociations from various rotational states of  $\text{H}_2\text{O}_2$ . The derived high pressure rate constants for HO recombination are also in quantitative agreement with experimental results. However, a more detailed analysis of experimental dissociation and recombination falloff curves appears necessary which will be elaborated in part III of this series.<sup>85</sup> Falloff extrapolations of the experiments should be based on the detailed strong collision falloff expressions derived from the present  $k(E, J)$  which fully account for rotational effects. Some additional weak collision broadening effects should also be considered while

only very minor non-statistical ones will have influenced the shape of the falloff curves. The available experimental results for recombination so far are only fragmentary and, in addition, contaminated by still uncertain contributions from the competing reaction  $\text{HO} + \text{HO} \rightarrow \text{H}_2\text{O} + \text{O}$ .<sup>17,18</sup> The experimental dissociation results showed deviations from limiting low pressure behaviour at unexpectedly low pressures.<sup>14</sup> The present work suggests that this is not due to non-statistical lifetime distributions but to markedly asymmetric strong collision falloff curves. Part III of this series<sup>85</sup> applies the present calculations in detail to thermal dissociation/recombination experiments and shows that the fragmentary experimental data can be combined into a complete set of falloff curves, allowing one also to separate contributions of the competing reactions  $\text{HO} + \text{HO} \rightarrow \text{H}_2\text{O}_2$  and  $\text{HO} + \text{HO} \rightarrow \text{H}_2\text{O} + \text{O}$ . The  $\text{H}_2\text{O}_2$ -system is then another example demonstrating the necessity and the opportunities of combining theoretical and experimental results for the construction of an internally consistent picture of the reaction dynamics.

## Appendix

### Molecular parameters used in modelling

(i) Parameters for  $\text{H}_2\text{O}_2$ .—Dissociation energy  $\Delta H_0^\circ = E_0$  with  $E_0/\text{hc} = 17052 \text{ cm}^{-1}$ ,<sup>31</sup> well depth of potential  $D/\text{hc} = 19039 \text{ cm}^{-1}$ . Rotational constants:  $A_e = 10.14 \text{ cm}^{-1}$  and  $B_e = 0.823 \text{ cm}^{-1}$ .<sup>45,58</sup> Harmonic frequencies in  $\text{cm}^{-1}$ : 3778, 1453, 889, 340, 3762, 1297<sup>45</sup> with the torsional frequency  $340 \text{ cm}^{-1}$  chosen such that the exact number of levels  $N(E)$  up to  $8000 \text{ cm}^{-1}$  is well reproduced, see text. Anharmonic zero-point energy:  $E_z/\text{hc} = 5690 \text{ cm}^{-1}$ .<sup>45</sup> Torsional levels in  $\text{cm}^{-1}$ : 0, 11.0, 259.3, 378.2, 792.2, 1023.8, 1267.3, 1513.9, 1761.4, 2005.4, 2262.4, 2424.3, 2767.2, ...<sup>43</sup>

(ii) Parameters for HO.—Rotational constant:  $B = 18.87 \text{ cm}^{-1}$ .<sup>45</sup> Vibrational levels in  $\text{cm}^{-1}$ : 0, 3571, 6975, 10216, ...<sup>45</sup> Fixed H–O distance:  $r_e = 0.963 \text{ \AA}$ .<sup>45</sup> Dipole moment  $\mu_D = 1.683 \text{ D}$ .<sup>45</sup>

(iii) Quantities derived for modelling of thermal dissociation/recombination rate constants.—Equilibrium constants  $K_c = [\text{HO}]^2/[\text{H}_2\text{O}_2] = 4.3 \times 10^{25} (T/1000 \text{ K})^{0.25} \exp(-24534 \text{ K}/T)$  molecule  $\text{cm}^{-3}$  over the range 1000–1500 K. Rotational factors:  $F_{\text{rot}} = 98, 56, 38, 27, 21, 17, 14, 12, 10, 9$ , and 8 for  $T = 200, \dots, 1200 \text{ K}$ .  $F_E = 1.03, 1.05, 1.07, 1.09, 1.10, 1.12, 1.14, 1.16, 1.18, 1.20$ , and 1.22 for  $T = 200, \dots, 1200 \text{ K}$ . Lennard-Jones parameters:  $\sigma/\text{\AA} = 4.196, 2.551, 3.542$ , and 3.798;  $\epsilon/k \text{ K} = 289.3, 10.22, 93.3, 71.4$  for  $\text{H}_2\text{O}_2$ , He, Ar, and  $\text{N}_2$ , resp.<sup>86</sup>  $F_{\text{anh}}(E_0) \approx 1.33$ ,  $a(E_0) = 0.948$ .

## Acknowledgements

Most fruitful discussions with D. Luckhaus on the question of numbers of vibrational states of  $\text{H}_2\text{O}_2$  are gratefully acknowledged. Thanks are also due to the Deutsche Forschungsgemeinschaft (SFB 357 “Molekulare Mechanismen unimolekularer Prozesse”) and EOARD (Grant award No FA8655-01-1-3034) for financial support.

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